

RACHEL TYPRIN

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ABOUT ME

Designer and ML developer building technologies and products that improve people's health, safety, and quality of life.

PERSONAL PROJECTS

MACHINE LEARNING DEVELOPER

SF Bay Area Microclimate Prediction

FEBRUARY 2026 - PRESENT

- Built a neighborhood-scale weather prediction model that delivers more precise local forecasts than regional models, improving severe weather predictions and enabling better emergency planning.
- Combined satellite data, geographic features, and hundreds of local weather station readings to train XGBoost and LSTM models that learn how hills, coastlines, and land cover shape weather at the neighborhood level, built end to end in Python on AWS S3.

PRODUCT PROTOTYPER AND DESIGNER

AI Prototyping Toolkit

FEBRUARY 2026 - PRESENT

- Developed an AI toolkit that Stanford PhD candidates and postdocs use to translate their research into real world solutions. Domains include substance abuse management and prevention, sustainable materials manufacturing, and urban development for community economic recovery in San Francisco's Tenderloin district.
- Enables rapid user research feedback, iterative product prototyping and low-cost development of MVP.

EXPERIENCE

RESEARCH ASSISTANT

Brain Connectivity Research

Drexel University

JUNE 2025 - PRESENT

- Identified which graph-based algorithms most accurately simulate brain connectivity, with direct applications to surgical planning and preserving cognitive and motor function during tumor removal.
- Validated findings by demonstrating correlations between simulated and real brain connectivity matrices, using Python for large-scale graph analysis and Matplotlib and Seaborn to communicate results.

FIRST AUTHOR

Human-Computer Interaction Research

Drexel University

JUNE 2024 - FEBRUARY 2026

- Pioneered a new research domain by defining and categorizing the "hopecore" cultural trend, producing the first structured dataset of its themes from social media and publishing findings as first author.
- Established a framework for analyzing how hopeful digital content spreads, with implications for designing platforms that actively support mental health.

CONTRACT PLAYTESTER

Video Game Playtesting

Video Game Marketing

DECEMBER 2022 - JUNE 2023

- Enhanced pre-release game quality by identifying usability issues, bugs, and glitches across multiple test builds.
- Delivered actionable feedback in team discussions, shaping user experience improvements prior to launch.

TEACHING AND LAB ASSISTANT

Computer Science Department Teaching Assistant

Oberlin College

SEPTEMBER 2022-DECEMBER 2022

- Supported 40+ students in an introductory CS course by clarifying core concepts, debugging code, and providing one-on-one guidance outside scheduled lab hours.
- Designed supplemental teaching exercises that strengthened student understanding of algorithms and programming fundamentals.

EDUCATION

DREXEL UNIVERSITY; BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

2023-2025

- A.J. Drexel Scholarship (merit-based)

OBERLIN COLLEGE; BACHELOR OF SCIENCE IN COMPUTER SCIENCE

2021-2022

- Dean's List, 2022, John F. Oberlin Scholarship (merit-based)

SKILLS

- Languages & Frameworks: Python, Java, JavaScript, SQL, LaTeX, C++, HTML, CSS
- Libraries and Tools: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PyTorch, NetworkX, Git, Docker, JetBrains IDE
- Concepts: Data Visualizations, Statistical Analysis, Machine Learning, Graph Algorithms, Human-Computer Interactions, Data Analysis, Search Algorithms (DFS, BFS, A*, heuristics), Software Architecture, Usability Testing, AI Prompt Engineering